

FRĪGEDÆG

GRAINS OVER A RECORDING

USER GUIDE · VERSION 1.0 · MAC

K.ODK

Grains over a recording

Frīgedæg runs **grains** over a recording. There is no synthesiser inside — everything you hear is a fragment of the material. Two reels and a ring of five cells keep the whole thing moving without a hand on it.

A test tone is loaded at the start, so it sounds at once. Open a file, or hold **MIC** to keep what the instrument itself is playing as the next recording, and the eight pads read it back as grains — where, how long, how high, how scattered, and through which filter and effect. Behind the pads sits **RENTHOUSE**: a ring of five slow oscillators and two reels driven by it.

Too much to read? Skip it. Leave MIXER… alone. Drag PLACE and GRAIN, step the key left of LEN a few times, and try FREEZE. The ring and the reels are there for when you want it to keep moving on its own.

In your DAW

The app, the AU (AUv3) and the VST3 are the same instrument. In your DAW you'll find **Frīgedæg** under *k.odk*; it plays on its own. As an audio effect, **Frīgedæg FX** takes what comes in as its recording (see page 9).

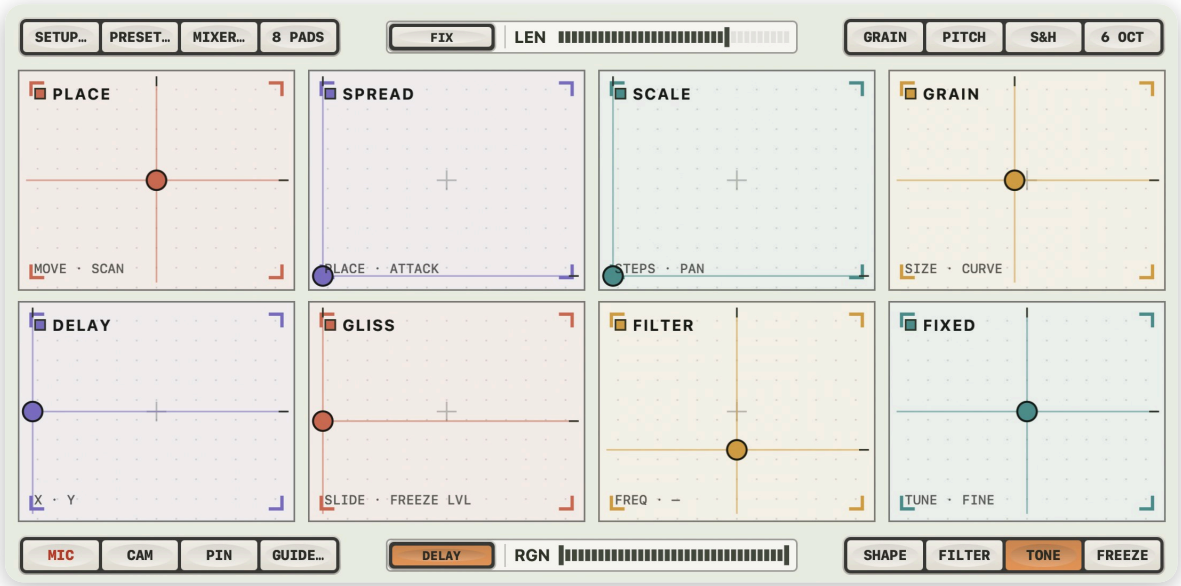
Installing

1. Open **Frīgedæg.pkg** and follow the installer. You can choose: **Standalone + AU**, and/or **VST3**.
2. Open **Frīgedæg** once from Applications — this registers the AUv3.
3. In your DAW, rescan plug-ins. Frīgedæg is under **k.odk**.

Standalone + AU	The Frīgedæg app in /Applications, with its Audio Unit (AUv3) inside — the instrument and the effect (Frīgedæg FX). Logic, GarageBand, Live and other AU hosts list them under *k.odk*.
VST3	In /Library/Audio/Plug-Ins/VST3 — Live, Bitwig, Reaper, Cubase …
The app	Plays on its own through the output chosen in the Mac's Sound settings.
Camera	Only for CAM. The first time, the Mac asks (in a DAW, for the DAW). The picture is read as movement and light; nothing is kept.

Requires macOS 12 or later · Apple silicon and Intel.

Two bars of keys, eight pads



Keys with "…" open a room; Done or Esc closes it. Keys with several names step through them on each click.

SETUP… PRESET… MIXER… 8 PADS	The desk · materials, camera, RANDOM, presets · RENTHOUSE, the ring and the reels · eight pads or four.
FIX LEN	How the grains are thrown, and LEN — the longest a grain may be.
GRAIN PITCH S&H 6 OCT	The engine · the pitch: PITCH, S&H, and how wide (6 / 8 / 10 OCT).
MIC CAM PIN GUIDE…	Hold MIC to keep what plays as the next recording · the camera moves the pads · the read point stays put · a short guide.
DELAY RGN	The effect, and RGN — the part of the recording that is read. Drag near either end to move that end.
SHAPE FILTER TONE FREEZE	Shaping the sound: the grain's shape, a filter on the grains, the colour on the way out, the reverb held.

Throwing the grains

FIX	An even stream from the read point. No random numbers.
SCATTER	Spacing and position scattered — a cloud. Moving fast bows it: more grains, shorter.
TOUCH	Each pad is a surface to strike: where you click sets that grain's length, pitch and place.
LFO · GATE	The ring in RENTHOUSE drives the grains · a grain each time the ring's gate opens.
RIPPLE	A click drops a seed that keeps throwing grains, spreading and fading.
LOOP	The two reels become a ring, and what comes out is fed back in as the next material.

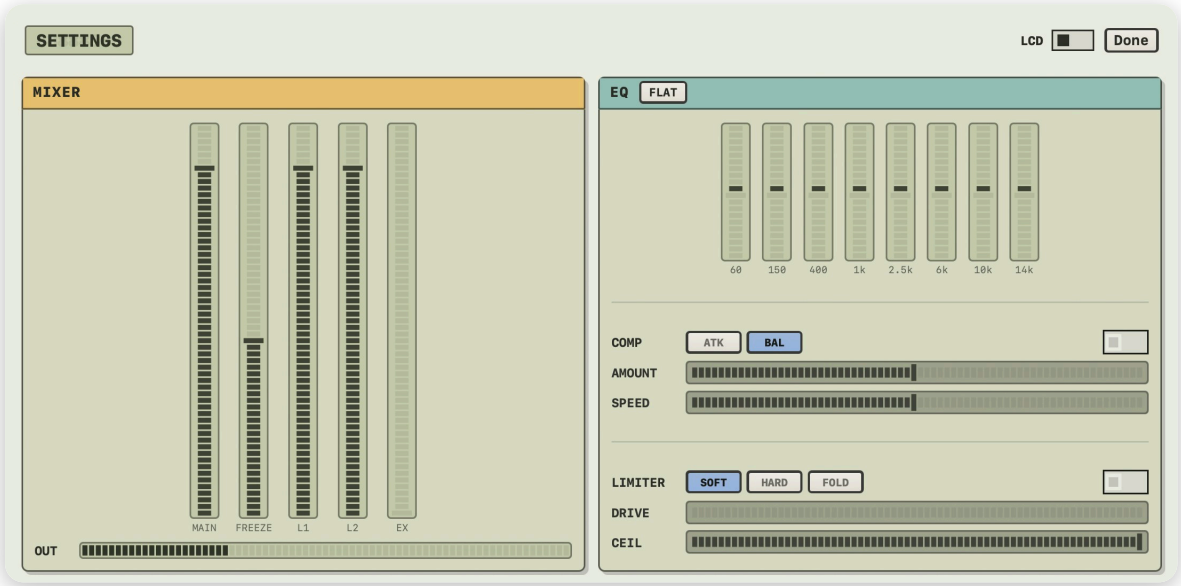
Engines, pitch, shape

GRAIN · PULSE · FEED	Plain overlap · pulsar synthesis: the rate is the pitch, the grain length the formant · reads a ring its own output is written back into.
PITCH · S&H	Each grain its own pitch from nine clean ratios, gliding inside the grain · a ratio held for 4 … 16 grains, then another.
6 / 8 / 10 OCT	The width of the pitch pad.
SHAPE	PERC (struck) · SUS (a held tone) · REV (every grain backwards).
FILTER	Only the grains pass through: LP, HP, BP, COMB, RING, SHIFT, CRUSH, STUT. The filter pad takes its two knobs.
TONE · FREEZE	A lift above 1 kHz and a soft saturation (TONE / RAW) · the reverb tail held, circling.
The effect	DELAY, VERB, HARM, FOLD, DIGI. Its pad is its two knobs.

Eight squares, or four

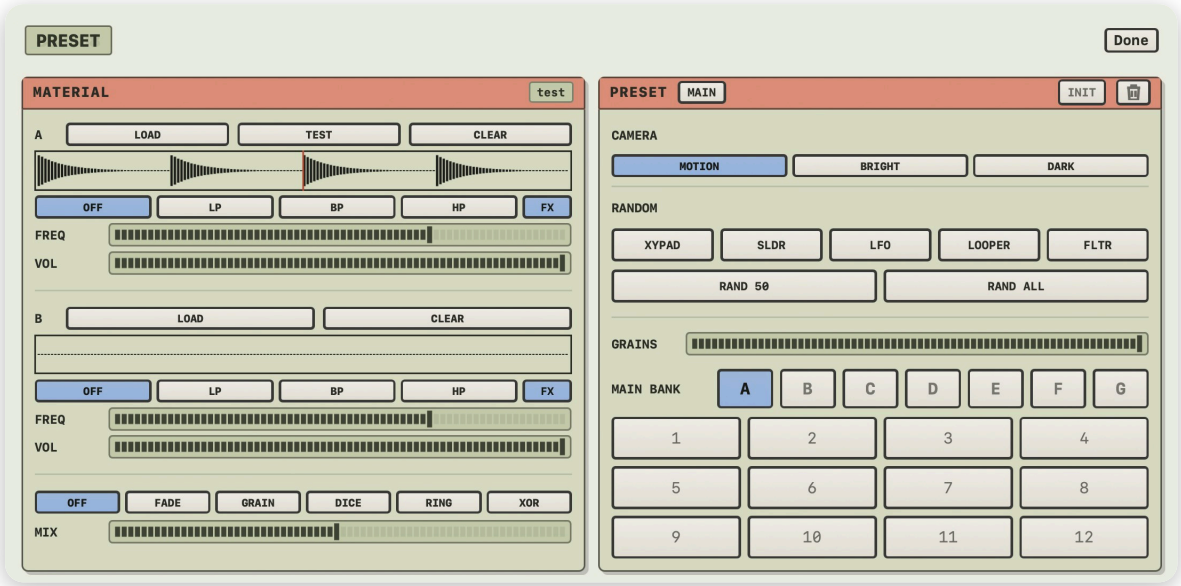
PLACE	X moves the read point, Y how fast it travels on its own (the middle stops).
SPREAD	X how far the grains scatter from the read point, Y the attack.
SCALE	X the steps: 1 to 8 pitches walked in order. Y the stereo spread.
GRAIN	X the grain length (the formant in PULSE; PULSAR), Y how the ramps bend.
The effect’s pad	Named after the effect — DELAY, VERB … — and its two knobs.
GLISS	X how far each grain slides, up or down. Y the FREEZE level.
The filter’s pad	Named after the FILTER key — LP, BP … — and its two knobs.
The mode’s pad	X the pitch (its width set by OCT), Y a fine trim. In LOOP: SIZE and RATE.
4 PADS	PLACE, GRAIN, the effect and the pitch, with a cross of sliders between them: the attack, the spread, the scale steps and the stereo spread.
Drag	Drag a point. With CAM on, the camera holds the points: each follows the movement, the brightest or the darkest of its square of the picture.

The desk



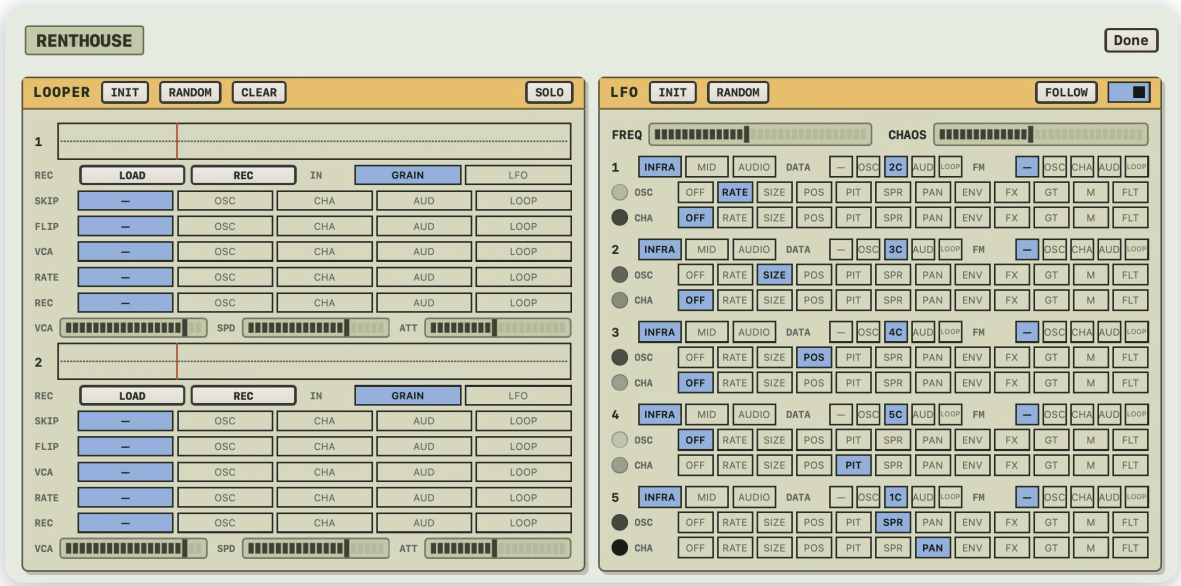
MIXER	MAIN, FREEZE, the two reels L1 and L2, EX (the ring's exit filter). OUT shows the level.
EQ · FLAT	Eight bands, 60 Hz … 14 kHz; FLAT puts them back.
COMP	ATK (the peaks, quick) or BAL (the average, slow), AMOUNT, SPEED. Its switch.
LIMITER	SOFT · HARD · FOLD, DRIVE, CEIL. Its switch.
LCD	The pads and the windows as an old handheld game's liquid crystal. The choice is kept.

Material and presets



MATERIAL A · B	LOAD a sound file (TEST: the test tone back into A), CLEAR. A filter for each — OFF, LP, BP, HP, FX — with FREQ and VOL. Up to two minutes each.
The mix of A and B	OFF, FADE, GRAIN, DICE, RING, XOR, and MIX.
CAMERA	What the pads follow: MOTION, BRIGHT or DARK.
RANDOM	Held, not clicked: XYPAD, SLDR, LFO, LOOPER, FLTR each redraw one corner; RAND 50 two or three of them, RAND ALL the lot. What moves goes there over a few seconds.
GRAINS	How many grains at once.
1 … 12 · A … G	Click to go back, hold to keep; the bin, then a slot, empties it. A … G are seven layers; the chip by PRESET switches MAIN and LIVE. INIT (held): everything back.

The ring and the reels



LFO	Five petals in a ring, each a triangle and two follow-and-hold circuits. FREQ, CHAOS. INIT, RANDOM, FOLLOW; the switch unplugs the whole ring, the patch stays.
INFRA · MID · AUDIO	Each petal's range. Up at AUDIO it becomes pitch.
DATA · FM	What steps who the petal chases (— OSC, a petal, AUD, LOOP) · what moves its speed.
OSC · CHA	The triangle and the battlement, each sent to a place: RATE, SIZE, POS, PIT, SPR, PAN, ENV, FX, GT (the gate), M (the ring), FLT (the exit filter). Nothing moves until one is chosen.
LOOPER 1 · 2	Two reels, always turning. LOAD a file or REC (a switch: a fresh loop, written until pressed again). IN: what is recorded — GRAIN or LFO. INIT, RANDOM, CLEAR, SOLO.
SKIP · FLIP · VCA · RATE · REC	Each takes one of the ring's outputs (— OSC, CHA, AUD, LOOP): a bookmark and back · the direction turned · the level · the speed · the grains written over what is read.
VCA · SPD · ATT	The reel's level, speed and how its gates open.

Frīgedæg FX, and a few ways in

On an audio track, **Frīgedæg FX** takes what comes in: it is written round a ring of two seconds, and that ring is the recording. Everything else is as the instrument — the pads, the modes, the engines, the effect, the filter, FREEZE, RENTHOUSE. Not there: LOAD, TEST, CLEAR, material B, MIC. With nothing coming in it falls silent; FREEZE holds.

A first sound	PRESET… › LOAD a field recording into A, then drag PLACE slowly and GRAIN to the left.
A cloud	Step the mode to SCATTER, raise SPREAD, turn PITCH on.
Hold it	FREEZE, then GLISS's Y for how loud the held tail is.
Let it run	MIXER… › LFO: set one petal's OSC to POS. The grains wander by themselves.
A reel as an oscillator	Patch a reel's own output into its SKIP: the fragment turns into a waveform, the gate speed into pitch.
Keep it	Hold a preset number. The song keeps its own state too.
In a DAW	The host sees eight knobs for automation — the pads, LEN, the filter and the level.

Credits

The ring and the gate-driven reels are influenced by instruments designed by Peter Blasser; nothing of his is reproduced. The granular vocabulary comes from Dennis Gabor, Iannis Xenakis and Curtis Roads.

Frīgedæg — k.odk (Kei Odaka) · keiodak.github.io/privacy/frigedaeg-support.html · kodk.dev@gmail.com

Requires macOS 12 or later · Apple silicon and Intel.